

Package ‘FirnR’

September 17, 2016

Title What the Package Does (one line, title case)

Version 0.0.0.9000

Description What the package does (one paragraph).

Depends R (>= 3.3.1)

License What license is it under?

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CIce	<i>Specific heat capacity of ice</i>
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Description

#Paterson 1994 copied from Goujon, C., J.-M. Barnola, and C. Ritz (2003), Modeling the densification of polar firn including heat diffusion: Application to close-off characteristics and gas isotopic fractionation for Antarctica and Greenland sites, J. Geophys. Res., 108(D24), 4792, doi:10.1029/2002JD003319.

Usage

CIce(T)

Arguments

T	ice temperature [K]
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Value

specific heat capacity of ice J/(kg K)

Author(s)

Thomas Laepple

DensityHL	<i>Herron Langway firn density</i>
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Description

Herron Langway firn density model, analytical solution

Usage

DensityHL(rho.surface, t.mean, bdot, depth = 0:9000/100)

Arguments

rho.surface	surface density in kg/m ³
t.mean	10 firn mean temperature in Kelvin
bdot	accumulation rate in kg/m ² /year
depth	snow depth in m

Details

based on the Matlab Code from Alan Grinsted #<https://www.mathworks.com/matlabcentral/fileexchange/47386-steady-state-snow-and-firn-density-model/content/densitymodel.m>

Value

list(depth, rho) with snow depth and density in kg/m³

Author(s)

Thomas Laepple examples result=DensityHL(rho.surface=340,t.mean=273.15-31.5,bdot=177,depth=0:150)
 plot(result\$rho,result\$depth,ylim=c(150,0),xlim=c(200,1000),xlab="firn density",ylab="depth (m)",type="l",lwd=2,mai
 simulated density")

DiffusionLength	<i>water isotope diffusion length in firn</i>
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Description

Diffusion length in firn

Usage

```
DiffusionLength(rho = NULL, z, dz, T, P = 0.65, b = 0.064, dD = FALSE)
```

Arguments

rho	firn density <i>*vector*</i> [kg/m ³]; if not provided, a linear density profile is used
z	firn depth <i>*vector*</i> [m]; observation points of 'rho', must be of same length as 'rho'
dz	depth resolution <i>*vector*</i> [m]; may vary over 'z', must be of same length as 'z'
T	temperature [K], either a scalar value or a vector of the same length a rho
P	surface pressure [atm]
b	annual mean accumulation rate m.w.e per year?
dD	<i>*logical*</i> ; if true, the diffusion length for d2H is returned, otherwise for d18O

Details

Thomas add details

Value

list(z=z,rho=rho,sigma=sigma), firn depth, density, diffusion length in cm

Author(s)

Thomas Muench modified by Thomas Laepple

Examples

```
depth<-0:150
t.mean<-273.15-30
bdot=200
rho<-DensityHL(rho.surface=340,t.mean=273.15-31.5,bdot=bdot,depth=depth)$rho
result.d018<-DiffusionLength(rho,depth,dz=1,T=t.mean,b=bdot/1000,dD=FALSE)
result.dD<-DiffusionLength(rho,depth,dz=1,T=t.mean,b=bdot/1000,dD=TRUE)
plot(result.d018$sigma,result.d018$z,ylim=c(150,0),xlim=c(0,12),xlab="diffusion length (cm)",ylab="depth (cm)",lwd=2,col="red")
lines(result.dD$sigma,result.dD$z,lwd=2,col="black")
legend("topleft",col=c("black","red"),lwd=2,c("d18O","dD"),bty="n")
```

Diffusivity

*Diffusivity of water isotopes in firn***Description**

Diffusivity of water isotopes in firn

Usage

```
Diffusivity(rho, T, P, dD = FALSE)
```

Arguments

rho	firn density [kg/m ³]
T	annual mean temperature [K]
P	surface pressure [atm]
dD	dD - *logical*; if true, the diffusivity for d2H is returned, otherwise for d18O

Details

Thomas: add details / references

Value

diffusivity in [cm²/s]

Author(s)

Thomas Muench modified by Thomas Laepple

Examples

```
T<-(-60:0)
plot(T,Diffusivity(rho=340,T=273.15+T,P=0.65,dD=FALSE),type="l",lwd=2,main="temperature dependence of diffusivity")
lines(T,Diffusivity(rho=340,T=273.15+T,P=0.65,dD=TRUE),col="red",lwd=2)
legend("topleft",col=c("black","red"),lwd=2,c("d18O","dD"))
```

FirnTemperature	<i>Simulate the Firn temperature</i>
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Description

Simulate the firn temperature based on the Fourier law of heat diffusion with an input signal given as the superposition of the annual cycle and its second harmonics

Usage

```
FirnTemperature(t, z, core = NULL, A1 = core$A1, A2 = core$A2,
  phi1 = core$phi1, phi2 = core$phi2, T0 = core$T0, kappa)
```

Arguments

t	time in years
z	snow depth in m, surface = 0, 1 = 1m deep
core	list of core parameters or alternatively NULL, in this case A1,A2 ... have to be given explicitly
A1	Amplitude of first harmonic (~1/2 of seasonal range)
A2	Amplitude of second harmonic
phi1	Phase of first harmonic
phi2	Phase of second harmonic
T0	Mean temperature in Kelvin
kappa	Thermal diffusivity of firn m ² /s

Value

vector of temperature at time t at the depth levels z

Author(s)

Thomas Laepple

Examples

```
plot(1:10,type="n",xlim=c(-45,-10),ylim=c(20,0),main="NGRIP")
for (iMonth in 1:12)
  lines(FirnTemperature(iMonth/12,(0:1000)/50,A1=16.5,A2=3,phi1=0,phi2=0,T0=-31.5,kappa=KappaFirn(273-31.5,rh
```

KappaFirn	<i>thermal diffusivity of firn</i>
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Usage

KappaFirn(T, rho)

Arguments

T	firn temperature [K]
rho	firn density [kg/m ³]

Value

thermal diffusivity of firn m²/s

Author(s)

Thomas Laepple

KappaIce	<i>thermal diffusivity of ice</i>
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Usage

KappaIce(T)

Arguments

T	ice temperature [K]
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Value

thermal diffusivity m²/s

Author(s)

Thomas Laepple

KFirn	<i>Thermal conductivity of firn</i>
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Description

Thermal conductivity of firn W/(m K) Schwander et al., 1997 copied from Goujon, C., J.-M. Barnola, and C. Ritz (2003), Modeling the densification of polar firn including heat diffusion: Application to close-off characteristics and gas isotopic fractionation for Antarctica and Greenland sites, J. Geophys. Res., 108(D24), 4792, doi:10.1029/2002JD003319.

Usage

KFirn(T, rho)

Arguments

T	firn temperature [K]
rho	firn density [kg/m ³]

Value

Thermal conductivity of firn W/(mK)

Author(s)

Thomas Laepple

References

Schwander et al., 1997

KIce	<i>Thermal conductivity of ice W/(mK)</i>
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Description

Thermal conductivity of ice W/(mK) Weller and Schwerdtfeger 1971 copied from Goujon, C., J.-M. Barnola, and C. Ritz (2003), Modeling the densification of polar firn including heat diffusion: Application to close-off characteristics and gas isotopic fractionation for Antarctica and Greenland sites, J. Geophys. Res., 108(D24), 4792, doi:10.1029/2002JD003319.

Usage

KIce(T)

Arguments

T	firn temperature [K]
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Value

Thermal conductivity of ice W/(m K)

Author(s)

Thomas Laepple

ParcelTemperature	<i>Temperature of a snow parcel</i>
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Description

Simulate temperature against depth for a snow parcel starting at a particular season based on the Fourier law of heat diffusion. This assumes a constant layer thickness, thus ignoring densification. As the temperature below 5m is close to constant, this approximation should be reasonable.

Usage

```
ParcelTemperature(startTime, depth, core)
```

Arguments

startTime	time in the year when the parcel started (years)
depth	depth vector (m)
core	list containing the core parameters (here A1,A2,phi1,phi2,T0,density and accum) are used

Value

vector of temperatures at the depths given by the depth vector

Author(s)

Thomas Laepple

Examples

```
kohnen<-list(lat=-75.00,lon=0,accum=72,density=345,T0=273.15-44.5,A1=16.7,A2=6.6,phi1=0,phi2=0,P=0.65,name=
depth<-seq(from=0,to=10,by=1/100)
plot(depth,ParcelTemperature(0.5,depth,kohnen),main="Temperature of a parcel at Kohnen",xlab="snow depth",)
```

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